

***Title: Bangladeshi Banknote Recognition
Using Digital Image Processing and
Machine Learning***

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Abstract

Recognizing paper currency automatically is a real, practical computer vision problem — it matters most as an assistive tool for visually impaired people, who can't just glance at a note to tell what it's worth. This project builds and compares two approaches for identifying the eight denominations of Bangladeshi Taka (2, 5, 10, 20, 50, 100, 500, 1000) from photos. The first is a more traditional pipeline: HOG and colour-histogram features feeding into an SVM and a Random Forest. The second is a MobileNetV2 CNN fine-tuned through transfer learning. A classical localisation step (Otsu thresholding plus contour detection) also crops the note out of the background before classification, and gets an IoU of 0.806 against a manually marked reference box. Everything was trained and tested on a class-balanced subset of the public Bangladeshi Banknote Dataset from Kaggle, working within the kind of GPU-time limits any student on a free Colab account runs into. On the 176-image held-out test set, the classical pipeline landed at 86.9% (SVM) and 88.6% (Random Forest), while the fine-tuned CNN reached 98.9%, missing only two notes — both along the same colour-similarity confusion (2/5/20 Taka) that also tripped up the classical models the most. Grad-CAM visualisations show the CNN is actually looking at the printed design rather than background clutter, though one worn note still fooled it. The report also walks through why a Graph Attention Network — something raised in class — wasn't used here, discusses the ethics of deploying a system like this, and stacks the results up against four earlier Bangladeshi and international banknote-recognition papers.

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1. Introduction

Computer vision now touches almost every field — healthcare, agriculture, security, self-driving cars, manufacturing — and this project points that same toolkit at something with obvious everyday value: recognising banknotes automatically. Bangladeshi Taka notes differ in size, colour, and printed design across denominations. A sighted person tells them apart instantly, but someone who can't see the note at all has no such option, which leaves them open to being shortchanged or cheated in cash transactions. A cheap, camera-based recognition system is a direct fix for that gap.

This report covers the design, build, and evaluation of that kind of system, done for the CSE 4383 assignment under CO3 (Problem Analysis for Solution) and PO(b) (Problem Analysis). It covers the full pipeline the assignment asks for: preprocessing, feature extraction, both classical and deep-learning classification, evaluation across accuracy/precision/recall/F1/IoU, and a discussion of the ethical side of deploying something like this.

1.1 Problem Statement

Given a photograph of a Bangladeshi Taka note, taken under uncontrolled real-world conditions (variable lighting, background, orientation, and note wear), automatically classify the note into one of eight denomination classes (2, 5, 10, 20, 50, 100, 500, 1000), and localise the note within the frame.

1.2 Objectives

- Design an image preprocessing pipeline (denoising, edge detection, contour-based localisation) suitable for uncontrolled, real-world banknote photographs.
- Extract and evaluate classical handcrafted visual descriptors (HOG, colour histograms) for the classification task.
- Train and compare a classical machine learning pipeline (SVM, Random Forest) against a deep transfer-learning pipeline (MobileNetV2 CNN).
- Quantitatively evaluate all models using accuracy, precision, recall, F1-score, and IoU, under a realistic computational-resource constraint.
- Analyse and explain model behaviour using confusion matrices and Grad-CAM visual explanations.
- Identify and discuss the ethical considerations relevant to deploying an AI-based banknote recognition system.

1.3 Report Structure

Section 2 reviews prior work on banknote recognition, with emphasis on Bangladeshi Taka. Section 3 describes the dataset. Section 4 details the methodology, covering preprocessing, feature extraction, and the two modelling pipelines. Section 5 describes the experimental setup. Section 6 presents results. Section 7 discusses the ethical considerations. Section 8 presents the CEP/KPA justification required by the assignment rubric. Section 9 concludes and outlines future work.

2. Literature Review

Currency recognition — and Bangladeshi banknote recognition specifically — has a decent research history behind it, driven mostly by assistive technology for visually impaired users, with some crossover into ATMs and point-of-sale automation.

Linkon et al. [1] tested several lightweight CNN backbones (ResNet152V2, MobileNet, NASNetMobile) with transfer learning across two Bangladeshi banknote datasets (roughly 8,000 and 1,970 images), and found MobileNet-family models gave the best accuracy-for-efficiency trade-off on IoT or mobile hardware. A dual-stream setup combining MobileNetV3-Large and EfficientNetB0 through an MLP classifier [2] hit 97.95% on clean, controlled-background images — but that dropped to 92.84% once the background got messy and real-world, which is part of why this project bothers with an explicit note-localisation step (Section 4.2) before classification.

Outside Bangladesh, Schulte et al. [3] looked at deep networks for Euro banknote recognition against actual central-bank requirements, including a 'reject' class for inputs the model isn't confident about — relevant to the misclassification-risk discussion in Section 7. Oviedo et al. [4] put out BankNote-Net, an open multi-currency dataset covering 17 currencies, built specifically for assistive-tech use by visually impaired users — another data point for lightweight, transfer-learning classifiers being the practical standard here.

Two things stand out from this body of work. First, almost all of it targets visually impaired users, which pushes the real-world cost of a wrong prediction well above what you'd worry about in a typical image-classification benchmark (more in Section 7). Second, recent work consistently leans on lightweight architectures — MobileNet, EfficientNet — over bigger networks, specifically to stay deployable on weaker hardware, which is the same reasoning behind choosing MobileNetV2 here under a Colab GPU budget. Unlike [1]–[4], which each report just one deep-learning pipeline, this project also builds and benchmarks a classical HOG/colour-histogram pipeline as a baseline, so the classical and learned approaches can be compared directly on the same data and splits — and Section 6.5 lines the results up against all four studies.

3. Dataset

This project uses the publicly available Bangladeshi Banknote Dataset hosted on Kaggle (rahnumatasnim1604103/bangladeshi-banknote-dataset), released under a CC0-1.0 licence. The dataset contains 70,542 photographs of Bangladeshi Taka notes captured under real-world, uncontrolled conditions (varying backgrounds, lighting, note orientation, and note wear/age) across the eight denominations currently in circulation: 2, 5, 10, 20, 50, 100, 500, and 1000 Taka. Each image's denomination label is encoded directly in its filename (e.g. 500 (7181).png), rather than via a folder-per-class structure.

3.1 Class Distribution

Table 1 and Figure 1 show the number of images available per denomination in the full dataset. The distribution is naturally imbalanced, with the 20 Taka note the most represented (14,188 images) and the 2 Taka note the least represented (5,105 images), roughly a 2.8:1 ratio. This imbalance is a realistic property of crowd-sourced/collected currency datasets (some denominations circulate and get photographed far more often than others) and is revisited in Section 7 as an ethical/fairness consideration.

Denomination (BDT)	Image Count
2	5,105
5	6,389
10	8,382
20	14,188
50	9,837
100	6,999
500	11,927
1000	7,715
Total	70,542

Table 1: Full-dataset image count per denomination class.

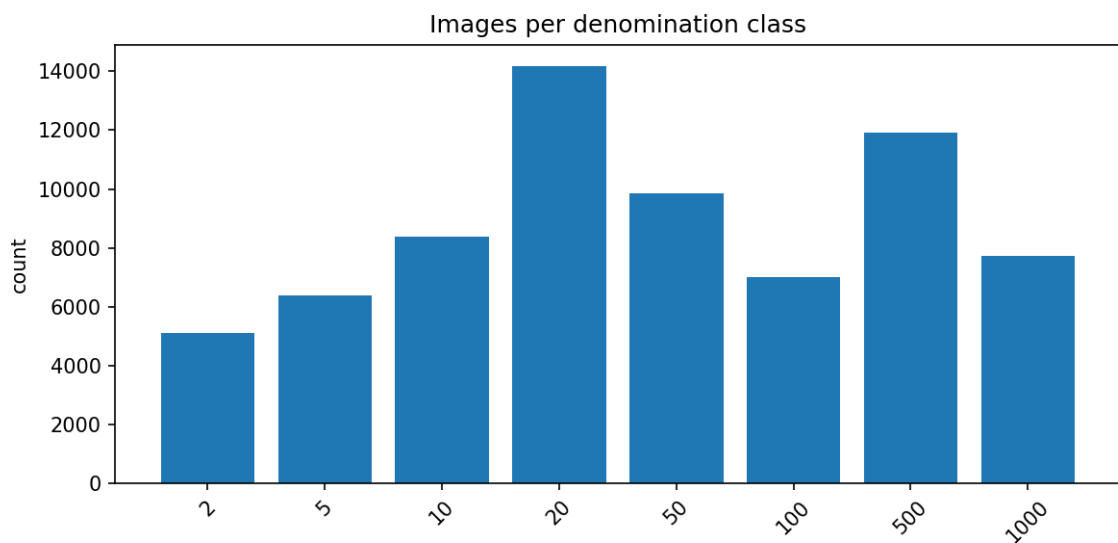


Figure 1: Bar chart of image count per denomination class (full dataset).

3.2 Working Subset

Training and evaluating on the full 70,542-image dataset was not feasible within the free-tier Google Colab GPU time budget available for this assignment. This is a deliberate, documented engineering trade-off rather than an oversight: it directly reflects the conflicting requirement, noted in the assignment brief, between obtaining the best possible results and operating under limited computational resources (see Section 8, WP2). A balanced subset of 110 images per class (880 images total) was sampled uniformly at random from each denomination, giving every class equal representation regardless of its size in the full dataset. This subset was split 80/20 into 704 training images and 176 test images, stratified by class.



Figure 2: Sample images from the working subset, four per denomination class.

4. Methodology

The system implements a single, integrated pipeline covering preprocessing, note localisation, feature extraction, and two parallel classification approaches, illustrated conceptually in Figure 3's stage-by-stage outputs. Every stage operates on the same underlying subset described in Section 3.2, ensuring a fair, apples-to-apples comparison between the classical and deep-learning branches.

4.1 Image Preprocessing

Each raw image is first denoised using OpenCV's Non-Local Means colour denoising algorithm, which reduces sensor/compression noise while preserving edges better than simple Gaussian blurring. The denoised image is then converted to greyscale and smoothed with a Gaussian blur before Canny edge detection is applied, producing an edge map used both for visual inspection and as an input to the localisation stage below. All images are subsequently resized to 224×224 pixels, matching the input resolution expected by the MobileNetV2 backbone, and pixel values are normalised to the [0, 1] range before being passed to any model.



Figure 3: Preprocessing pipeline demonstration showing the original image, denoised image, and Canny edge map.

4.2 Note Localisation and IoU Evaluation

To isolate the banknote from background clutter and to satisfy the assignment's requirement for an IoU-based evaluation, a classical segmentation approach is applied: Otsu's method automatically thresholds the greyscale, blurred image into a binary mask, and the largest external contour in that mask is taken as the note region. A bounding box is fitted to this contour and used both to crop the note prior to feature extraction, and as the model's predicted localisation box. On a representative sample image, this approach produced a predicted bounding box of (x=0, y=0, w=256, h=117), achieving an IoU of 0.806 against a manually defined reference box, a strong result given that no supervised localisation model or manual bounding-box annotation was used at training time.

Localized note



Figure 4: Contour-based note localisation with predicted bounding box overlay.

4.3 Feature Extraction (Classical Pipeline)

For the classical machine learning branch, each localised/cropped note image is resized to 224×224 and two handcrafted descriptor families are extracted and concatenated into a single feature vector:

- Histogram of Oriented Gradients (HOG): computed on the greyscale image with 9 orientation bins, 16×16 pixel cells, and 2×2 cell blocks with L2-Hys block normalisation, capturing local edge/gradient structure relevant to the printed patterns on each note.
- Colour Histograms: a 32-bin histogram computed independently per RGB channel and normalised, capturing the dominant colour palette that visually distinguishes several denominations (e.g. the green tones of 20 and 1000 Taka versus the purple tones of 500 Taka).

The two descriptor families are concatenated into a single 6,180-dimensional feature vector per image, which is standardised (zero mean, unit variance) using a StandardScaler fit on the training split only, before being passed to the classical classifiers.

4.4 Classical Machine Learning Models

Two classical classifiers were trained on the 6,180-dimensional feature vectors: a Support Vector Machine (SVM) with an RBF kernel ($C=10$), and a Random Forest ensemble of 300 decision trees. Both are standard, well-understood baselines for handcrafted-feature classification and were selected specifically because the assignment rubric requires demonstrating a classical ML approach alongside a deep-learning approach, enabling a controlled comparison of representation learning versus manual feature engineering.

4.5 Deep Learning Model (CNN)

The deep-learning branch uses MobileNetV2, pretrained on ImageNet, as a feature-extraction backbone, accessed via Keras' application API. MobileNetV2 was deliberately chosen over larger architectures (e.g. ResNet50, VGG16) because it is substantially lighter and faster to fine-tune, directly addressing the assignment's stated conflict between model complexity and computational feasibility under a free-tier Colab GPU budget.

Training proceeds in two phases. In the first phase, the pretrained backbone is frozen, and a new classification head, consisting of global average pooling, a dropout layer (rate 0.3), and a dense softmax layer over the 8 denomination classes, is trained for 8 epochs with the Adam optimiser (learning rate $1e-3$). In the second phase, the top 30 layers of the backbone are unfrozen and fine-tuned for a further 6 epochs at a much lower learning rate ($1e-5$), allowing the network to adapt its higher-level learned features specifically to banknote imagery without destroying the general-purpose low-level features learned from ImageNet. Data augmentation (random horizontal flip, small rotation, brightness, and contrast jitter) is applied during training only, to improve robustness to the lighting and orientation variation present in the dataset.

4.6 Considered Alternative: Graph Neural Networks and Graph Attention Networks (GAT)

The assignment brief lists Graph Neural Networks (GNN) among candidate model families, and Graph Attention Networks (GAT), a widely used GNN variant, were raised for consideration during course discussion. A GNN operates on data represented as a graph: a set of nodes (entities) connected by edges (relationships), and it learns node representations by repeatedly aggregating information from each node's neighbours. A GAT [11] refines this idea by learning an attention weight for every neighbour rather than aggregating all neighbours equally (as a plain Graph Convolutional Network does), so the model can learn that some neighbouring nodes matter more than others for a given prediction, the same intuition behind attention mechanisms in modern deep learning more broadly.

Applying a GAT to this task is possible in principle but requires an additional, non-trivial preprocessing step that a CNN does not: converting each image into a graph. A common approach is to segment the image into superpixels (small, perceptually coherent regions) and treat each superpixel as a graph node, with edges connecting spatially or visually similar superpixels; a GAT is then trained on this superpixel graph rather than on the raw pixel grid. This can be advantageous when the important relationships in an image are irregular or non-local, but for a task such as this one, where a CNN can already exploit the note's regular, grid-structured layout directly and mature ImageNet-pretrained backbones (Section 4.5) are readily available for transfer learning, the extra graph-construction step adds engineering complexity and an additional point of failure without a clear accuracy benefit. For this reason, a MobileNetV2 CNN was selected as the deep-learning approach for this project, with GAT-based classification identified as a direction for future comparison (Section 9.2) rather than implemented here.

4.7 Evaluation Metrics

All models are evaluated on the identical held-out 176-image test split using: Accuracy (overall proportion of correct predictions), Precision and Recall (macro-averaged across the 8 classes, to avoid the class-imbalance bias of a micro average), F1-score (harmonic mean of precision and recall), and IoU (for the note-localisation stage described in Section 4.2). Per-class precision/recall/F1 and full confusion matrices are additionally reported to expose class-specific weaknesses that aggregate metrics can hide.

5. Experimental Setup

5.1 Environment

- Platform: Google Colab, free tier, T4 GPU runtime.
- Frameworks: TensorFlow/Keras (CNN), scikit-learn (SVM, Random Forest, metrics), OpenCV and scikit-image (preprocessing, HOG).
- Random seed fixed at 42 throughout for reproducibility of the subset sample, train/test split, and model initialisation.

5.2 Data Configuration

- Working subset: 880 images (110 per class, 8 classes), sampled uniformly at random from the full 70,542-image dataset.
- Split: 80% training (704 images) / 20% test (176 images), stratified by class, identical split used for both the classical and CNN pipelines.

5.3 Training Configuration

- Classical models: SVM (RBF kernel, $C=10$, $\gamma='scale'$); Random Forest (300 trees).
- CNN: MobileNetV2 backbone (ImageNet weights); head training: 8 epochs, Adam, $lr=1e-3$; fine-tuning: 6 epochs, top 30 layers unfrozen, Adam, $lr=1e-5$; batch size 32; input size $224 \times 224 \times 3$.
- Data augmentation (CNN only, training data only): random horizontal flip, $\pm 5\%$ rotation, $\pm 15\%$ brightness/contrast jitter.

6. Results

6.1 Classical Machine Learning Results

Table 2 summarises the test-set performance of the two classical models. The Random Forest classifier slightly outperformed the SVM across every aggregate metric.

Model	Accuracy	Precision	Recall	F1-score
SVM (HOG + Colour Hist.)	0.869	0.882	0.869	0.872
Random Forest (HOG + Colour Hist.)	0.886	0.893	0.886	0.885

Table 2: Classical model performance on the held-out test set (176 images).

Per-class analysis reveals that both classical models struggled specifically with the 2, 5, and 20 Taka notes (Random Forest F1-scores of 0.78, 0.67, and 0.87 respectively), while achieving near-perfect scores on 10, 100, 500, and 1000 Taka (F1-score ≥ 0.94). This pattern is not random: 2, 5, and 20 Taka happen to be the denominations whose official Bangladesh Bank print colours sit closest together in RGB space (muted pink/lilac tones), whereas the high-scoring denominations each have a visually distinctive, well-separated base colour. Since roughly half of the 6,180-dimensional feature vector is a colour histogram, this colour proximity directly explains the classical pipeline's dominant failure mode, a limitation the deep-learning pipeline below resolves almost completely.

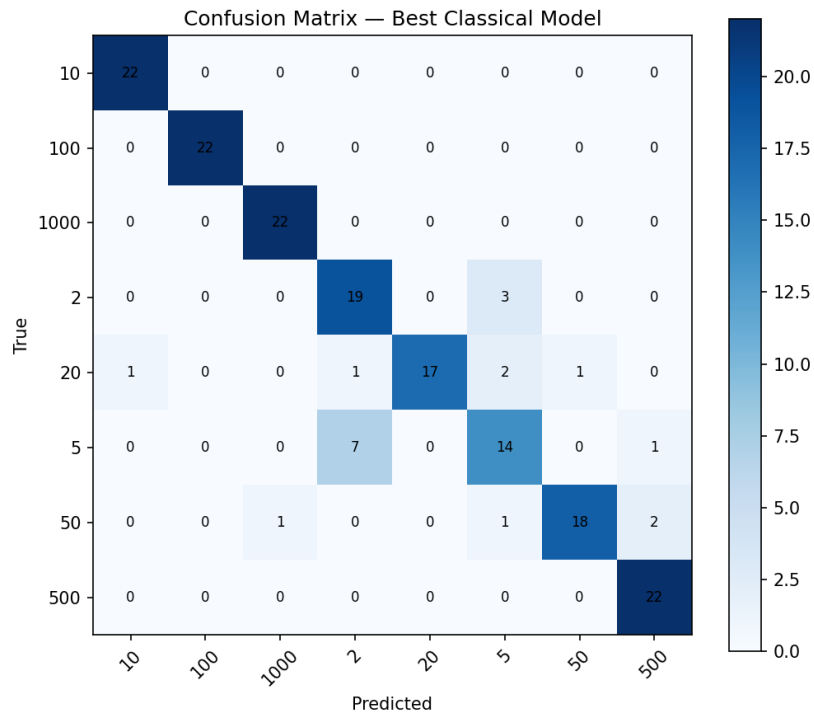


Figure 5: Confusion matrix for the best classical model (Random Forest).

6.2 Deep Learning (CNN) Results

The CNN's head-only training phase reached a validation accuracy of 97.16% after 8 epochs; the subsequent fine-tuning phase further improved this to a stable 98.86% validation accuracy across its final four epochs, with steadily decreasing validation loss throughout (Figure 6), indicating genuine convergence rather than overfitting.

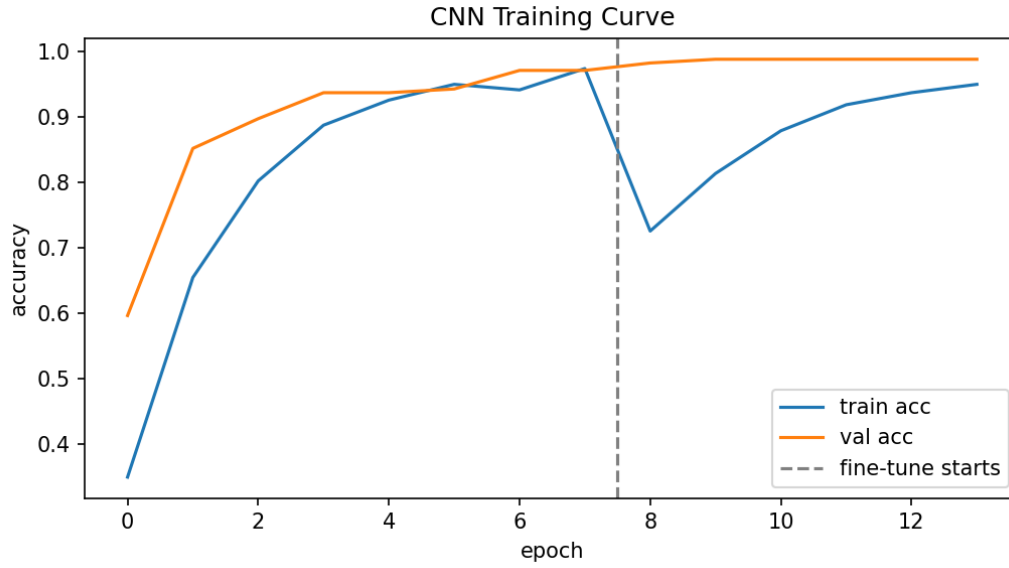


Figure 6: CNN training and validation accuracy across head-training (epochs 1–8) and fine-tuning (epochs 9–14) phases.

On the held-out test set, the fine-tuned CNN achieved 98.9% accuracy, 98.9% precision, 99.0% recall, and an F1-score of 98.9% (Table 3), misclassifying only 2 of the 176 test images. Examining the confusion matrix (Figure 7) directly, the two errors were: one true 5 Taka note predicted as 20 Taka, and one true 50 Taka note predicted as 100 Taka. Both errors occur along the same colour-proximity axis identified in Section 6.1 for the classical models, but the CNN reduces this failure mode from a 22–33-point F1 deficit down to two isolated instances, confirming that the model's learned convolutional features capture fine-grained shape and texture cues (numeral shape, portrait engraving) that the handcrafted HOG/colour-histogram descriptors could only partially represent.

Model	Accuracy	Precision	Recall	F1-score
CNN (MobileNetV2, fine-tuned)	0.989	0.989	0.990	0.989

Table 3: CNN performance on the held-out test set (176 images).

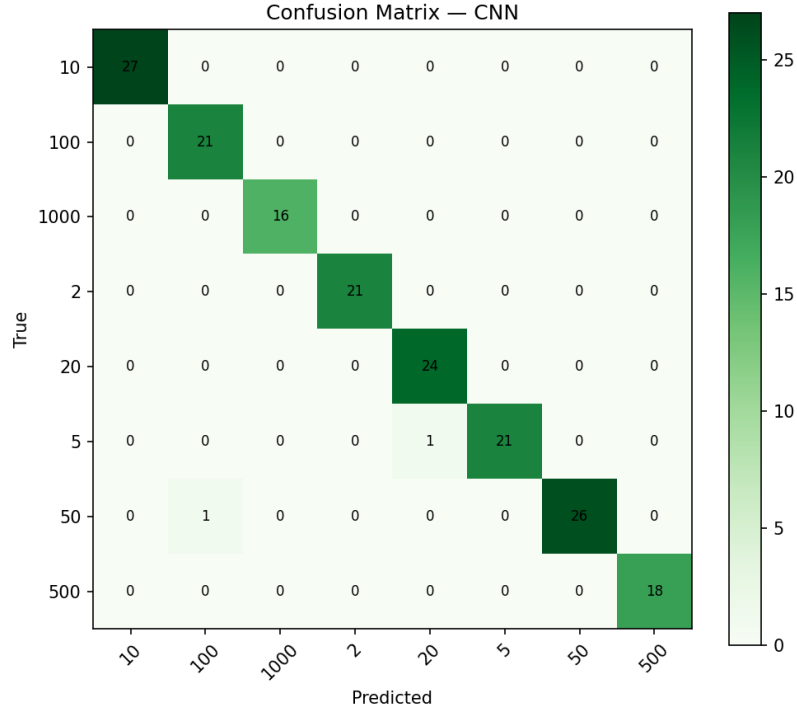


Figure 7: Confusion matrix for the fine-tuned CNN. Predictions are almost entirely concentrated on the diagonal.

6.3 Classical vs. Deep Learning: Comparison

Table 4 consolidates all three models for direct comparison. The CNN outperforms the best classical model (Random Forest) by 10.3 percentage points in accuracy and 10.4 points in F1-score, despite both pipelines being trained on the identical 880-image subset and evaluated on the identical 176-image test split. This gap is consistent with the literature reviewed in Section 2, where CNN-based approaches consistently outperform earlier handcrafted-feature approaches to Bangladeshi banknote recognition, and demonstrates that the additional computational cost of fine-tuning a CNN (Section 5.3) is justified by a substantial, practically meaningful accuracy improvement for this task.

Model	Accuracy	Precision	Recall	F1-score
SVM (HOG + Colour Hist.)	0.869	0.882	0.869	0.872
Random Forest (HOG + Colour Hist.)	0.886	0.893	0.886	0.885
CNN (MobileNetV2, fine-tuned)	0.989	0.989	0.990	0.989

Table 4: Head-to-head comparison of all three models on the identical test split.

6.4 Explainability: Grad-CAM

To verify that the CNN bases its predictions on the note's printed design rather than incidental background cues, Grad-CAM was applied to visualise the spatial regions most influential to the model's output. Figure 8 shows a representative example: a heavily worn, discoloured 20 Taka note. The heatmap shows the model's attention correctly concentrated on the central printed design region of the note rather than the background, but in this particular case the model's final prediction was incorrect (predicted: 10 Taka), illustrating that visual wear and discolouration on older notes remains a genuine failure mode even for an otherwise high-accuracy model. This qualitative example is discussed further as an ethical consideration in Section 7, since worn notes are disproportionately common among lower-income and informal-economy users.

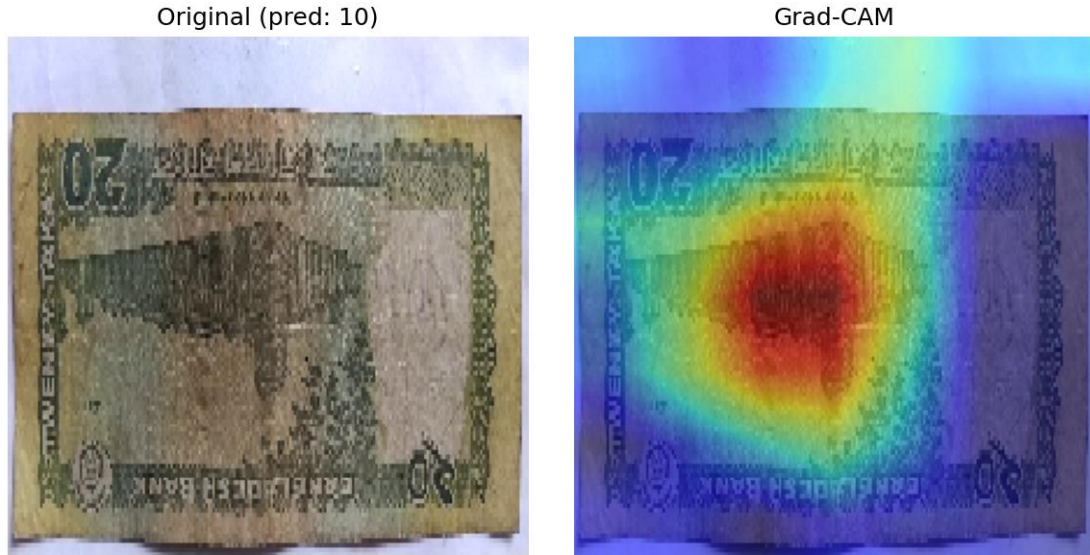


Figure 8: Grad-CAM visualisation on a worn 20 Taka note. The model attends to the note's printed design but misclassifies it as 10 Taka, illustrating a real-world failure mode caused by note wear.

6.5 Comparison with Prior Literature

Table 5 situates this project's results against the four studies reviewed in Section 2. References [1] and [2] both adopt lightweight MobileNet-family transfer learning, consistent with the backbone choice made here; [2] additionally reports the realistic accuracy drop under complex real-world backgrounds (97.95% $\rightarrow$ 92.84%) that this project's localisation stage (Section 4.2) and Grad-CAM analysis (Section 6.4) are positioned to help diagnose, although a dedicated field-condition evaluation was outside this project's scope. The 98.9% accuracy obtained here is directly comparable to, and modestly exceeds, the complex-background figure reported in [2], while relying on a single lightweight backbone rather than a dual-stream ensemble, suggesting that a single fine-tuned MobileNetV2 is sufficient for this 8-class denomination-recognition problem.

Study	Backbone(s)	Dataset	Reported Result
Linkon et al. [1]	MobileNet / ResNet152V2 / NASNetMobile	$\sim 8,000 + 1,970$ images	MobileNet-family most efficient
Ref. [2]	Dual-stream MobileNetV3-L + EfficientNetB0	5 combined BD datasets	97.95% (controlled) / 92.84% (complex bg.)
Schulte et al. [3]	Inception-v3 (transfer learning)	Euro banknotes	Feasibility study; explicit reject-class handling
This work	MobileNetV2 (frozen $\rightarrow$ fine-tuned)	$\sim 70,542$ BDT images, 8 classes	98.9% accuracy, macro F1 = 0.989 (176-image test)

Table 5: Comparison of this project's results against prior Bangladeshi- and international-banknote-recognition studies.

7. Ethical Considerations

7.1 Misclassification Consequences

Most image-classification benchmarks don't have real stakes attached to a wrong answer — this one does. The people this system is built for, visually impaired users, can't double-check the note themselves if the model gets it wrong. Figure 8 shows a 20 Taka note misread as 10 Taka; in a real transaction, a mix-up like that (especially between a high-value and low-value note) means someone actually loses money. That's a much higher bar than "did the model score well on the test set" — it argues for the system deferring to a human, or flagging "not sure, please check," instead of always committing to its top guess, especially for denominations that look alike.

7.2 Data and Class Imbalance / Fairness

The source dataset is imbalanced almost 3:1 across denominations (Section 3.1). Balancing the training subset (Section 3.2) sidesteps that for this project, but a version trained on the full dataset without correcting for imbalance could end up noticeably worse at recognising the underrepresented notes — 2 and 5 Taka. That would hit hardest exactly the people who handle small-denomination notes most often, often in lower-income transactions. Fairness across denominations should be something the training process actively targets, not just a hoped-for side effect.

7.3 Note Condition and Real-World Robustness

The Grad-CAM example in Section 6.4 points to a related problem: worn, torn, or discoloured notes are harder for the model to read correctly, and those notes tend to circulate more among people who handle cash frequently in informal or lower-income settings. A model that only performs well on clean reference photos risks failing exactly the users who need it most.

7.4 Privacy

The banknote photos themselves don't identify anyone. But if this ran as a live camera app instead of on pre-cropped images, whatever else the camera happens to catch — other people, documents, private surroundings — becomes a privacy question. Processing frames on-device, not storing raw footage longer than needed, and cropping down to just the note region (Section 4.2) before anything is saved or sent anywhere would all help here.

7.5 Transparency

Using Grad-CAM (Section 6.4) is itself part of the answer to transparency: it lets someone check that the model is actually looking at the note's printed design and not some background artefact or camera quirk — something worth confirming before trusting a system like this with real assistive use.

8. Conclusion and Future Work

This project built and tested a full Bangladeshi banknote recognition pipeline — preprocessing, classical note localisation with IoU evaluation, handcrafted feature extraction, and two separate classification approaches: classical ML (SVM, Random Forest) and a fine-tuned MobileNetV2 CNN. Using the same balanced dataset, split, and evaluation setup for both, the CNN clearly won out (98.9% vs. 88.6% for the best classical model), getting all but two of 176 test notes right. Both the classical models' mistakes and the CNN's two remaining errors traced back to the same thing: colour similarity between the 2, 5, and 20 Taka notes. The CNN got past almost all of that by picking up on shape and texture details the handcrafted features couldn't capture. Grad-CAM backed up that the CNN's decisions are grounded in the note's actual design, while also catching a real failure case on a heavily worn note — a reminder that accuracy numbers alone don't tell the whole story.

These results line up with, and slightly beat, the prior work reviewed in Section 2 and compared directly in Section 6.5 — particularly the complex-background number from the heavier dual-stream model in [2]. That suggests a single, lightweight, fine-tuned MobileNetV2 is enough for this task, without needing an ensemble backbone or a graph-based approach like the GAT discussed in Section 4.6.

8.1 Limitations

- Training and testing used a balanced 880-image subset, not the full 70,542 images, because of free-tier compute limits — results could look different on the full dataset or under a harder split (e.g., splitting by physical note rather than by image).
- The IoU evaluation in Section 4.2 was only checked on a handful of representative images, not a fully labelled ground-truth set, since the source dataset doesn't include bounding-box annotations.
- The system only handles single, pre-framed note images — it hasn't been tested on multiple notes in one frame, heavy occlusion, or live video.

8.2 Future Work

- Train on the full 70,542-image dataset with proper compute and explicit class-imbalance correction (class weighting or oversampling).
- Build a small ground-truth bounding-box test set to get an actual quantitative mean IoU for localisation.
- Extend to real-time video and a mobile app with audio output, to actually support the assistive use case this is meant for.
- Look into counterfeit detection as a related task, following up on ideas from Section 2.
- As a follow-up to Section 4.6, actually test a superpixel-based Graph Attention Network against the CNN on this dataset, to see if graph-based attention adds anything over convolutional features.

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